

Pando: One App to Unite, Inform and Mobilize

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Introduction

Social media's role in activism is highly contested. While it can serve as a useful tool for disseminating information and updates to international audiences, it fails to create the same change as in person mobilization (Kidd & McIntosh, 2016).

In fact, it often enables the ability for users to passively engage with environmental movements, believing that the likes and comments they share online equate to real mobilization. This is one of many gaps we have identified in current social media platforms, alongside the endless streams of information and promotions they supply to users and the lack of moderation applied to their content. In 2022, climate-sceptic tweets reached an all-time high of 850,000 posts, increasing significantly from the 220,000 shared in 2020 (Calma, 2022). This signalled an urgent need for a social media platform that not only provides users with accurate and timely information regarding the climate crisis and its associated movements, but also one that helps facilitate on-the-ground activism.

We designed Pando as a response to this growing need. This social media platform, named after the world's largest tree, aims to educate users on the climate crisis and steps they can take to create positive change in their local and global communities. Our mission with Pando is to create a space for individuals to share and access accurate information about the environment, while inspiring environmental activism across local and global communities. Our vision is to foster an inclusive and engaged community of well-informed climate activists that engage with environmental justice in a variety of ways. This includes anyone around the world ages 14 and up.

Many values guided the design of Pando to ensure the platform achieved our above-stated goals. Our leading value was education, around which we designed the app's two primary features: the Hope Hub and news feed. These features support the sharing of information and personal

anecdotes to promote education between users. They also facilitate collaboration among activists and foster hope by highlighting ways people are tackling the climate crisis. Accessibility is another guiding factor which aims to make environmental knowledge more comprehensible and facilitate global connections across movements, regardless of users' background, age, language, etc. Our final value is action, as Pando's main goal is to encourage local and global mobilization that aims to better the health of our planet and all its living creatures.

With Pando's Hope Hub and news feed, we plan to facilitate proper education on the climate crisis and the mobilization of environmental activism, promoting action amongst individuals from local to global scales.

Background

We have chosen the name Pando for our platform, which roughly translates to "I spread" in Latin. The initial inspiration for the name came from Pando, an ancient quaking aspen tree or clone colony, which is the largest tree in the world and considered to be the heaviest living organism. It is a colony of identical quaking aspen stems, which originate from a single root underground. The connectivity of the tree's clones through their root is emblematic of our hope to facilitate connectivity and collaboration across temporal and spatial boundaries.

Pando is believed to have grown during the last ice age, approximately 80,000 years ago. The age and continued growth of the organism in the face of changing climate and its environment are a testament to its resilience, which connects to our value of hope. All agitation is resilience, and to continue the environmental movement, all of us need to be resilient.

Taking into account its translation, “I spread”, we hope this platform is successful in raising awareness about the climate crisis and further spreading the environmental movement to increase awareness and activism globally.

Functions and Features

There are many components comprising Pando’s functions and capabilities. Primarily, information sharing on the app will take place in two different discussion forums. This is different from other social media platforms, as many provide users with one feed to share and intake factual news alongside personal stories. As a result, it can be difficult for people to differentiate between the two forms of information and become untrustworthy of the information they see online. Therefore, we have decided to separate information sharing into one dominated by discussions of accurate environmental policies and climate news, and the other used to share users’ personal sustainability journeys to inspire and learn from others.

We have called the latter forum the Hope Hub, a place where all success stories and solutions can be posted. This is an essential component of the platform, as climate content often focuses only on the impending doom of the universe, leading to disengagement. This feature of Pando will spotlight current climate victories to hopefully spark hope, increasing online and real-life engagement in the users.

Although raising awareness is a vital component of stimulating activism, getting involved with climate action is daunting. Pando offers users a variety of features to facilitate in person mobilization. These tools include petitions, event maps, volunteering opportunities, community projects and more. Activists can plan climate strikes, clean-ups, or protests in the app and those who want to partake can RSVP through the same platform. Furthermore, Pando will have weekly/monthly goals set out for users to complete both virtually and in-person. These challenges

will be provided by the app creators with help from experts in the field of sustainability and climate justice. Upon completion, users will earn a virtual tree to be “planted” on their home screen that can be viewed by other users. This feature aims to increase engagement by rewarding individuals who partake in local and global sustainability activities and climate events. Many people are aware of the climate crisis but lack the understanding on how to get involved in climate action. By integrating activism tools directly into the platform and allowing users to track their own sustainability progress, Pando makes it easier for users to engage with climate action and their immediate community.

Additionally, the subscription feature on the app will allow those to subscribe to or follow content relating to the user’s specific environmental interests. This can also help to narrow information and news to those pertaining to the user’s specific location. Information regarding climate change can be complicated and overwhelming, so being able to tailor the experience to specific interests that are unique to each person will help curate the content so each person can enjoy the app to its fullest.

Finally, arguably the most important capability and feature of this platform would be the fact-checking component. Misinformation is a major issue in the environmental world, which in turn will lead to inaction. Thus, users will be encouraged to cite sources when sharing anything other than personal experiences. Additionally, moderation features will be implemented both to verify facts shared online and authenticate in-person mobilization efforts to ensure the safety of users.

Values

Education

Education is the one of the guiding values behind Pando. Social media is a powerful tool for disseminating information regarding climate news and political events that are subject to censorship on mainstream platforms. However, both mainstream media and social media can also be used to spread misinformation on these topics, making it difficult for users to differentiate between true and false narratives. Pando aims to educate the public while promoting the sharing of factual information by providing its users with two coinciding platforms - one to share information about global climate events and environmental activism and the other for individuals to share their own experiences and sustainability tips. This educates users on a range of scales, from distant global events to individual actions that impact their daily lives.

One feature of Pando allows users to share facts and news stories globally, making topics more accessible to people around the world. In her text about poverty coverage in the news, Redden (2011) discusses how mass media limits discussions of public issues by only sharing particular perspectives, skewing people's understanding of what issues, like poverty, really mean. Rather than highlight the opinions of those who control the media platforms, we were inspired to showcase multiple sightlines and a plurality of voices relating to climate issues (Estes & Dhillon, 2019). Through this, Pando aims to provide individuals with perspectives that other platforms neglect to include and give users a fuller understanding of critical topics relating to environmentalism and civil rights.

Pando also supports users educating other users through sharing their own experiences and tips to living more sustainably. This feature, the Hope Hub, also connects to value 4, collaboration, and value 5, empowerment. Sharing personal experiences related to sustainable living helps create an open dialogue between users to learn from one another and what tips work best for them,

forming a sense of community within the app. Furthermore, seeing other people adopt simple lifestyle swaps that help the environment can empower others to take steps in their own lives.

Additionally, users can subscribe to or follow specific topics that they find most interesting. This will serve to narrow the amount of information individuals see on the app. While this seems counterproductive in the world of education, it will ultimately prevent users from being overwhelmed by vast amounts of information coming from across the world. Information overload as a result of poor information quality and excessive information sharing through social media can lead to digital distrust or hopelessness, preventing users from taking action against climate-related issues (Fan et al., 2021). Therefore, limiting the amount of information shown to users can increase the quality of their understanding and ensure they trust the information they are receiving.

Most popular social media platforms and internet forums are controlled by wealthy conglomerates in positions of power, often with close ties to political figures. Kidd and McIntosh (2016) raise concerns over who makes, controls and has access to those platforms, and question what information they allow to be shared. Since purchasing Twitter, Elon Musk has provided platforms to racists and conspiracy theorists with no attempt to prevent their spread of misinformation and hateful rhetoric (Solnit, 2024). These emphasize the need for people, particularly civil rights and environmental activists, to have access to Pando to educate others and learn the truth for themselves, unfettered by the agendas of tech elites and political figures.

Hope

Hope is a fundamental component of Pando, designed to counteract the feeling of despair often associated with climate change and leading to inaction. This platform will emphasize optimism and resilience and show real progress to inspire individuals to participate in climate action. We hope to accomplish this through four main components.

Firstly, one of the primary features of Pando is the amplification of success stories, solutions, and progress in environmental justice movements. The “Hope Hub” is a discussion forum where users can go to find and contribute real-world stories of environmental wins, ranging from personal lifestyle victories to policy changes at the governmental level. This draws on Estes’ (2019) text on the #NoDAPL movement, which showcased resistance against the construction of the Dakota Access Pipeline while simultaneously reaffirming indigenous sovereignty and strengthening alliances across communities. Stories such as this emphasize the importance of resistance in creating meaningful change, thus reinforcing that collective action can truly lead to victories. Additionally, The Whole Earth Catalog, as discussed by Turner (2006), described how its purpose was to provide individuals with tools to change their world. Using this as inspiration, we intend for Pando to be a digital resource for hopeful environmental solutions.

Unlike most social platforms that prioritize passive scrolling and content consumption, Pando will promote hope by encouraging participation through interactive components. There will be opportunities to create and participate in petitions, organize events within the community, and collaborate on climate projects. This way, users will be able to feel as though they are a part of real change and, in turn, provide hope for others. Inspired by Kavada's (2015) research on social movements, Pando will be a place to connect users with local initiatives and try to foster action within communities.

Most importantly, Pando will also create a network where users can connect to others who share encouragement and collaborate on climate initiatives. Using Kidd and McIntosh's (2016) model of community-driven digital spaces, the platform will reflect this by providing a space for those who are new to the environmental movement to connect with seasoned activists. Connecting

with others who share the same ambitions will help foster these feelings and transition them into meaningful actions.

Finally, the platform will shift the idea of the climate crisis from an uncontrollable situation to an approachable issue. While acknowledging the urgency and need for climate action, Pando will frame the content and posts around actionable solutions through the citing and verification of the posts. Redden (2011) highlights that the media often sensationalizes the climate crisis and yet doesn't provide meaningful solutions that people can partake in. Pando plans to counter all of these sensationalized headlines by highlighting solutions and success stories.

Overall, having hope as one of Pando's core values will help users feel empowered to contribute to climate action. It is important as fostering hope is an essential component for sustaining real activism and change.

Accessibility

Pando prioritizes accessibility, ensuring that all users have access to environmental knowledge and actionable insights, regardless of socioeconomic class, physical ability, or geographic location. This principle is critical for breaking down the obstacles that limit effective participation in digital activism.

Many digital platforms inherently favour those who are digitally literate and have consistent internet access, leaving marginalized populations at a disadvantage (Hjorth & Hinton, 2019). To address this digital divide, Pando is designed to function efficiently on low-bandwidth connections and offers a text-heavy version for users with limited data availability. This approach aligns with the advocacy for community-driven technological infrastructures that serve the public good rather than being controlled by monopolistic tech corporations (Barendregt et al., 2021). By

providing an alternative to ad-supported sites, Pando aims to democratize environmental discourse so that even those with minimal access to the internet can join the conversation.

Additionally, ensuring accessibility for users with disabilities is essential to Pando's mission. The platform follows established accessibility guidelines, such as the Web Content Accessibility Guidelines (WCAG), which provide a framework for making digital spaces more inclusive (W3C, 2023). The study emphasizes the necessity of incorporating accessibility from the development stage, rather than treating it as an afterthought. (Research, 2024). Pando will incorporate assistive technologies such as screen reader compatibility, keyboard navigation, and customizable text features to accommodate a diverse range of users.

Moreover, censorship is an essential concern Pando tries to tackle. Environmentalists are often monitored, with information censored, and content removed by state or corporate censorship in most regions (Earl et al., 2022). Such bans are focused in communities already experiencing environmental injustices (Brown & Younes, 2023). Pando does accept this form of control, instead embracing decentralized and open-source platforms that facilitate information sharing, storage, and retrieval without users relying on centralized and censorship-infested infrastructures (Shilina, 2023). By doing so, it supports the notion that environmental wisdom is not a privilege, but a right—making grassroots climate activism possible even within online speech-restrictive settings.

In addition to digital accessibility, linguistic inclusivity is also a consideration. Environmentalism is a global movement, but most platforms operate primarily in English, unconsciously excluding non-English-speaking activists' participation. Research indicates that English-language dominance in digital activism can exclude key perspectives and create barriers for those whose primary language is not English, affecting both engagement and representation (Ginsburg, 2024; Hjorth & Hinton, 2019). Pando broadens its reach by incorporating multilingual

support and fosters cross-cultural dialogue on climate issues. Multilingual platforms significantly enhance engagement and collective action, especially in advocacy movements that require widespread participation (Schradie, 2018).

Accessibility is not only a feature of Pando but a core component that guides its design and functioning. By addressing digital injustices, the use of inclusive design principles, and linguistic accessibility, Pando facilitates a more diverse population to engage in climate activism.

Collaboration

Pando seeks to create a space where environmentally conscious or curious people can come together and communicate on topics relating to environmental justice. The proposed platform is a place where people can collaborate and take action on problems they are passionate about, and where one can learn and streamline their knowledge absorption. Therefore, the value of collaboration is significant to this social media platform.

As previously mentioned, Pando will host two forums: one where users can post and share information, research, news, campaigns, etc. about the environment or narrower aspects of it like biodiversity conservation, climate change, environmental racism, etc. The other forum will allow users to share more personal stories, actions and tips concerning the environmental movement and sustainability, facilitating deeper connections with others. The former is designed to allow people with similar concerns to learn together, and in instances where possible, take action together. With the option of following smaller interests and aspects of the environmental movement, users can focus their attention on one issue. This lets like-minded individuals collaborate on multiple levels, from simple information sharing to signing petitions and even organizing. In the latter forum, users can post their own stories, and personal actions relating to the environmental movement. This will

allow interpersonal connection and collaboration outside of a community of users with similar interests.

Collective problems require collective solutions and such solutions require collaboration. The environmental movement aligns with many movements that seek to disrupt society's status quo and challenge the current capitalistic and anthropomorphic paradigm. Brown (2015) described how the neoliberal ideology seeks to alienate us from our communities by commercialising people. It instills the values of free market capitalism in all aspects, thus making us what Brown calls “homo oeconomicus” (Brown, 2015). We aim to build a platform antithetical to such values, one where collective action is possible and encouraged. The importance of collaboration is relevant in movements like the previously discussed #NoDAPL movement at Standing Rock. The indigenous people saw that the location of the pipeline endangered the lives of many human and non-human beings and rallied against it (Tallbear, 2019). Tallbear (2019) also points out how the movement was a culmination of the indigenous struggle against colonialism and represented more than a united front against a pipeline. This highlights the importance of collaboration across spatial and temporal boundaries in the movement. Additionally, the connectivity of the #NoDAPL movement to the #IdleNoMore movement in Canada (Tallbear, 2019) also demonstrates the connection between various environmental movements across the globe. Collaboration among these movements leads to increasing awareness, more discontent and increasing collective bargaining power. The successful management of any movement, especially on social media, is rooted in collaboration among individuals.

The concept of virality is rooted in the act of interaction and collaboration. Kavada's (2019) research into the Occupy Wall Street movement, more specifically their social media presence,

emphasizes the importance of collaboration. Active as well as passive participants on social media were important in disseminating the movement's message (Kavada, 2019).

Action/Empowerment

In today's digital world, social media has become a crucial force in shaping how people organize, advocate, and engage in climate justice. These platforms have transformed advocacy from a passive information-sharing model into one where people can interact, collaborate, and take direct action toward sustainability. Unlike traditional media, which often limits public participation, action-oriented platforms can empower individuals and communities to be active agents of change.

Social media now plays a pivotal role in environmental activism by offering tools for mobilization, education, and empowerment. According to Hart et al. (2024), "Facebook posts with an affective component—either positive or threat-based—are most likely to be associated with greater levels of engagement". This suggests that social media is more than a space for discussion; it's a catalyst for real-world impact. By emphasizing emotional resonance, platforms can turn passive viewers into climate advocates.

A major strength of social media lies in its ability to foster collective identity and unified action. Kavada (2015) explains that during the Occupy Movement, digital platforms enabled activists to form a shared identity in which "Occupy activists came to constitute themselves as a collective actor and the role of social media in this process". In the same way, climate justice campaigns can use digital tools to strengthen solidarity and rally people around urgent environmental issues.

Strategic communication is key to successful digital advocacy. Jackson (2019) argues that movements must create alternative media spaces to challenge mainstream narratives and empower marginalized voices. The author states that "[a]lternative media created by movements is a well-

documented strategy, as mass dissemination of activist messaging is crucial to promulgation and base-building efforts (Jackson 2019). These insights underscore the need for platforms that facilitate thoughtful and action-driven dialogue.

As such, our platform Pando, will incorporate these insights into its design. Key features include:

- Interactive challenges and gamification to incentivize sustainable practices.
- Clear calls to action, including petitions, policy advocacy, and community-led initiatives.
- Moderation and fact-checking tools to combat misinformation, ensuring reliable sources are prioritized in discussions.

Critics often argue that online activism stays confined to digital spaces, lacking real-world outcomes. To address this, Pando will bridge online and offline efforts by offering:

- A dynamic event map for local climate actions like protests and clean-ups.
- Volunteer opportunities with grassroots organizations.
- Collaborative tools for community leaders to co-create and share sustainable initiatives.

Ultimately, Pando is guided by the belief that social media should do more than inform; It should empower. By creating a platform that values inclusivity, encourages tangible action, and supports real-world impact, we aim to redefine how people participate in the climate movement. Through community engagement and innovative tools, Pando hopes to be a platform where climate action becomes not just possible, but inevitable.

Logistics

Moderation

In line with our app's values regarding education, we plan to moderate the activity that occurs on Pando. Calma (2022) highlights the consequences of leaving social media platforms unmonitored through an analysis of Twitter activity following its purchase by Elon Musk. Tweets including climate misinformation increased exponentially from 220,000 in 2020 to 850,000 in 2022, while a search for #climate reveals #climatescam as a top result (Calma, 2022). Without moderation of social media platforms, which rely on user-based content and participation, harmful and false information can easily spread. These ideas can easily gain traction as individuals rarely do external research regarding what they see on social media, therefore, Pando will do this extra work to ensure they are properly informed.

Moderating content on Pando aims to reduce the spread of misinformation and harmful rhetoric that is prevalent on most social media platforms. To make this moderation easier, we will present users with a code of conduct to which their posts should adhere. We have outlined some of these rules below:

1. Pando prohibits the use of discriminatory language towards users of any ethnicity, nationality, race, gender, sexual orientation, etc.
2. All information shared regarding events or facts must be proven by scientific or other means. Any sharing of known misinformation or disinformation will not be tolerated.
 - a. Users must provide/be able to provide necessary proof to support their claims.
3. All posts and content shared must relate to one or more of the following topics: environmental justice, civil rights activism, climate events.

We will provide the option for Pando users to report content they believe to be in violation of the code of conduct. We will then have human reviewers go through the flagged content and take action where appropriate. Moderators can choose to leave the post online as is, remove it from view, or in extreme cases, ban users from the app. Due to the young age of some of Pando's users, we will enforce stricter moderation on posts viewed by those under 18 years old. Along with the rules outlined in the code of conduct, any graphic content will be hidden from the feed of younger users.

Furthermore, one of Pando's core values is to empower action, with many of its features dedicated to facilitating mobilization efforts. To ensure the safety of our users, we will use similar moderation tactics to verify the legitimacy of any in-person activities. All individuals or groups who wish to organize an activity must submit an application stating some personal information, such as name and contact information, as well as details on the type of event they are looking to organize. These applications will first be vetted by the same algorithm used for posts, followed by verification from a human moderator. Only events that are approved by both processes will be posted to the Hope Hub for other users to see. For both moderation processes, users will have the option to appeal the decision made on their post, aligning with the open dialogue Pando prioritizes.

Moderating the activity on Pando is essential in maintaining a safe environment for all its users and preventing individuals from sharing harmful content on its platform. It will also help to preserve the credibility of the app and the trust of its users. As the name represents, Pando aims to be a platform that further connects activists in the environmental movement. Creating a platform that people can trust and feel safe in sharing information is a crucial step in growing the environmental justice movement.

Funding Model

The most sustainable and ethically aligned way to fund Pando is through grants and public ownership. This model would maintain Pando's independence, uphold its climate justice mission, and safeguard it from the pressures of advertising-driven revenue systems. Social media platforms such as Facebook and YouTube have historically transitioned toward monetizing user activity (data commodities and platforms) into ad-centric ecosystems. Pando, if driven by profit motives, risks falling into similar patterns and straying from its mission and values.

Thus, adopting a grant-funded and publicly-owned structure would help Pando avoid compromising its values for revenue. Truby (2020) emphasizes the risks of unchecked digital growth, noting that “unregulated roll-out of experimental AI poses risks to the achievement of the UN Sustainable Development Goals (SDGs)”. This illustrates the broader issue of profit-driven platforms undermining equitable and sustainable progress. In contrast, Pando could remain transparent, accountable, and mission-focused by sourcing financial support from governmental agencies, environmental NGOs, and organizations committed to sustainability and justice.

Public ownership also promotes participatory governance. Decisions could be made collectively with users, ensuring the platform remains reflective of its community's environmental and social justice priorities. Unlike venture capital-funded platforms that prioritize returns, a publicly-owned Pando could prioritize ethical design, user education, and community resilience. It would foster an online network of climate dialogue that is inclusive, non-exploitative, and free from corporate manipulation.

In conclusion, a grant-funded and publicly-owned approach is the most fitting for Pando's vision. It preserves its integrity, supports democratic participation, and avoids the pitfalls of

commercialized social media models. This funding strategy not only aligns with Pando's mission but strengthens its role as a transformative platform for climate justice.

Challenges and Barriers

One of the largest obstacles Pando will face is maintaining the validity of information relayed on the site while ensuring the site remains open and interactive as a dialogue space. Social media platforms have grappled with misinformation, particularly on environmental issues, as false claims have the ability to spread globally instantaneously (Hjorth & Hinton, 2019). To prevent this, Pando will include a fact-checking process that runs through user submissions, allowing users to vet facts and flag dubious posts. In addition, we will request users to include source citations for factual claims and provide transparency markers, such as credibility ratings for forwarded articles. An algorithm software will assist in detecting false information, but human moderators will make the final decisions on posts to avoid algorithmic biases.

Another primary challenge is encouraging active user engagement while realizing significant participation beyond scrolling. Most online platforms struggle to balance user-friendliness with more comprehensive forms of activism (Schradié, 2018). Pando will counter this by including gamified engagement elements such as sustainability challenges, discussion leaderboards, and rewards for verified engagement. These engagement features will motivate users to actively contribute while advancing the site's mission of environmentalism.

In addition, accessibility of the platform is a key concern. As research shows, the majority of advocacy platforms are English-oriented, which excludes non-English-speaking activists from actively engaging in online activism (Ginsburg, 2024). In response, Pando will enable multilingual

access through translation functions and discussion forums based on language, ensuring cross-cultural environmental discourse.

Sustainability of funding is also a risk. Since Pando is grant-funded and government-owned, securing long-term funding without corporate or advertising control will be critical (Barendregt et al., 2021). To mitigate this risk, Pando will seek partnerships with NGOs, universities, and environmental organizations, leveraging crowdfunding campaigns and public grants to maintain autonomy and stay mission-focused.

Lastly, moderation and maintaining an on-platform environment of respect will be necessary, as climate issues have a tendency to polarize. Pando will employ moderation tools to mark down misinformation, hate speech, and spam and encourage constructive debate. However, we acknowledge algorithmic bias and censorious by-products, particularly against marginalized voices in international environmental justice movements (Tufekci, 2017). In doing this, Pando will integrate automation and community flagging systems with open appeal processes to make sure that moderation is inclusive, equitable, and context-specific.

Pando will largely suppress state or corporate censorship, especially where environmental activism is made a crime in oppressive areas. The site reduces reliance on censorship-hungry websites with open-source and decentralized technology. It facilitates free sending and receiving of vital climate information (Shilina, 2023). Such technology makes grassroots action and global climate justice campaigns possible without stifling dissent.

Through active combat of these issues through open policies, community-driven interaction, linguistic inclusivity, and eco-friendly financial support, Pando will be assured success as an inclusive and fact-based environmental activist platform. Namely, Pando draws inspiration

from Indigenous-led movements, like #NoDAPL, that refuse to back down against the profit-driven systems that are degrading our planet's health (Brown, 2019). In doing so, Pando seeks to motivate a new wave of activists who can advocate for environmental sustainability and social justice.

Conclusion

We identified the rise of misinformation and disinformation in mainstream social media platforms and the need for effective climate action as two areas we wanted to address with Pando. It represents values which we deem important for climate justice and environmental movements. With education, hope, accessibility, collaboration and action in mind, we have devised this platform to allow all people to engage with these movements. We hope the dual forum allows people to learn, engage and organise in a safer and better informed environment, while connecting more personally with like-minded individuals. By learning about and sharing personal experiences and global stories, we seek to build a safe space for beginners and enthusiasts alike.

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